SARRTHI – Project Synopsis

A Proactive Compliance and Payment-Monitoring Platform for MSME Suppliers to Government Transport Bodies

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1. Introduction

Micro, Small and Medium Enterprises (MSMEs) that supply components, castings, forgings and fabricated parts to Indian Railways and other government transport bodies are protected by Sections 15 and 16 of the MSMED Act, 2006. The Act requires the buyer to pay within 45 days of acceptance of goods or services, and delayed amounts attract compound interest at three times the RBI bank rate. In practice, vendors discover a breach only after their working capital is already locked up.

SARRTHI is a web application that moves this discovery to the day the invoice is raised. It gives the vendor a running view of the 45-day clock, an estimate of the interest a delay may be worth, the buyer's payment behaviour and the evidence needed for a later claim.

2. Problem Statement

A vendor supplying a government transport body has no independent, self-serve system to:

1. see in real time whether an invoice is approaching or has crossed the 45-day window;
2. judge whether a buyer is a reliable payer before bidding on its next tender; and
3. keep orders, delivery proofs, acceptance records and correspondence organised so that a later claim can be supported.

The government's MSME ODR Portal (which replaced the Samadhaan Portal for new cases from 15 October 2025) starts only after a default has occurred, so nothing helps the vendor before that point.

3. Objectives

- Track every invoice against a configurable monitoring window (45 days by default) and show days remaining or overdue.
- Estimate interest on delayed payments and always show the assumptions behind each figure.
- Record part and full payments and keep balances and statuses consistent.
- Manage tenders, purchase orders, delivery and acceptance dates in one workspace.
- Provide a compliance view: procurement quota indicator, checklist, buyer delay report and a rule-based pre-bid risk score.
- Keep a document vault with an evidence checklist and an escalation timeline.
- Present a dashboard and exportable reports, including a working-capital estimate.
- Run end to end in a demo mode with realistic sample data and no server.
- Keep every user's data separate, and never present the product as an official portal or as legal advice.

4. Scope

In scope: vendor-facing workflows for a single vendor business, namely invoice and payment monitoring, orders and tenders, compliance intelligence, document vault, dashboard and reports.

Out of scope: arbitration, conciliation, mediation and legal advice (these belong to the MSME ODR Portal and the Facilitation Councils), direct integration with IREPS or GeM, live WhatsApp delivery, OCR invoice capture, crowd-sourced analytics and lender-facing APIs. The disclaimer "Academic Project — Not an Official Government Portal" appears on every screen.

5. Existing System and Gap Analysis

The MSME Samadhaan Portal (launched 30 October 2017) and the ODR Portal (launched 27 June 2025) handle disputes after a default. Reported figures show the scale of the problem: delayed payments owed to MSMEs of about ₹7.34 lakh crore (FY24), and about 81% of Samadhaan applications receiving no buyer response. Government sources name Railways, electricity and aeronautics among the sectors owing MSMEs the most.

SARRTHI addresses six gaps that remain after the 2025 upgrade:

1. no proactive tracking of a vendor's own invoices against the 45-day window;
2. no evidence base for raising a quota grievance;
3. no comparative view of buyer payment reliability before bidding;
4. no link between a vendor's order and invoice records and the dispute system;
5. no cash-flow view of what a delay costs; and
6. no coverage for medium enterprises, which the ODR mechanism excludes.

SARRTHI is a complement to the ODR Portal, not a substitute. Its figures are estimates, not guaranteed recoveries.

6. Proposed System and Modules

The Software Requirements Specification defines 50 functional requirements across nine modules. The current build covers the following.

Module	Main features
User, Identity and Access	Business profile (Udyam, GSTIN, PAN), per-user data isolation, sign-in via Internet Identity
Invoice and Payment Monitoring	Payment clock per invoice, part and full payments, what-if interest calculator, search, filter, purchase orders, working-capital estimate
Compliance Intelligence	25% procurement quota indicator, compliance checklist, buyer delay report, rule-based pre-bid risk score, tender tracking
Communication and Escalation	In-app alerts for upcoming and overdue deadlines, escalation steps

Module	Main features
Document and Data Capture	Document vault with categories, evidence checklist
Reporting and Dashboard	KPIs, monthly financials, buyer risk, CSV and print exports, activity feed
Demo mode	Browser-resident sample dataset with reset, no server needed

Modules for financial support (for example TReDS referral), collective intelligence and monetisation are deferred to a later phase by design.

7. Key Technical Features

- **Monitoring window:** starts from the acceptance date, falls back to the invoice date, and applies only to MSME-applicable invoices from covered buyers.
- **Interest estimator:** exact integer arithmetic using balance × rate × days ÷ 36,500, with assumptions printed beside every figure.
- **Pre-bid risk score:** weighted blend of checklist gaps (40%), bid size (30%) and the buyer's average payment delay (30%).
- **Data-source layer:** one flag switches the interface between the browser demo dataset and a REST client prepared for a future Spring Boot service.
- **Money handling:** whole-rupee integers (Nat in Motoko, bigint in TypeScript), never floating point.

8. Methodology

The team followed an incremental, document-led process over five weeks: project plan and market notes (Week 2), problem statement and feasibility report (Week 3), requirement analysis and use cases (Week 4), and the first draft of the Software Requirements Specification (Week 5, 7 August 2026). Implementation then proceeded module by module, with design decisions recorded in a design brief and project guidance file kept in the repository.

9. Technology Stack

Layer	Technology
Frontend	React 19, TypeScript, Vite, Tailwind CSS, shadcn/ui (Radix UI)
Routing and data	TanStack Router, TanStack Query, react-hook-form
Charts	Recharts
Backend	Motoko canister on the Internet Computer (38 domain endpoints)
Authentication	Internet Identity (passwordless)
Demo storage	localStorage-backed store with seed data
Testing	Vitest and Testing Library (14 frontend test files), PocketIC for backend tests
Tooling	pnpm, Biome, mops, Git

The specification calls for a Java 17 and Spring Boot backend with PostgreSQL and JWT. The frontend already contains the REST client, endpoint modules and proxy for it. The Java service itself is planned as a migration, not delivered code.

10. Current Status and Limitations

Against the 50 specified requirements, the build meets 17 in full, 15 in part and 18 not yet, and all 16 Must-priority requirements are at least partly met. Known limitations:

- only simple interest is implemented, although statutory compound interest is required;
- several backend date calculations need correction;
- the document vault stores metadata only, not file bytes;
- escalation steps and the evidence checklist exist in the backend but are not yet shown in the interface;
- settings such as consent toggles and password change are not persisted; and
- the Java and Spring Boot backend exists only as an integration seam and migration plan.

11. Expected Outcomes

- A working, mobile-responsive web application demonstrable without any server.
- Early warning on invoices before the 45-day window is breached, with transparent interest estimates.
- Better bidding decisions through buyer-level delay data and a pre-bid risk score.
- A well-organised evidence base that supports a later ODR filing.
- A documented requirements-to-code traceability record and a migration plan to Spring Boot.

12. Future Scope

- Statutory compound interest and correction of the backend date logic.
- A Java 17 and Spring Boot backend with PostgreSQL, Spring Security and JWT.
- Day-30 and day-40 reminders by WhatsApp or email, and pre-filled ODR filing data.
- Real file storage for the document vault, and Udyam verification.
- OCR invoice capture, Tally and GST e-invoice import, and PDF or Excel export.
- TReDS financier referral, crowd-sourced analytics and a rights chatbot with multi-language support.

13. References

1. Micro, Small and Medium Enterprises Development (MSMED) Act, 2006, Sections 15 and 16.
2. Ministry of MSME, S.O. 1364(E) dated 21 March 2025 (MSME classification limits).
3. MSME Online Dispute Resolution Portal, odr.msme.gov.in.
4. MSME Samadhaan Portal data (archive).
5. Income Tax Act, Section 43B(h), inserted by the Finance Act, 2023.
6. SARRTHI Software Requirements Specification (Week 5 draft) and SARRTHI Final Project Report, 2026.